

CHELYABINSK MEZHDUNARODNYY, Russian Federation

WMO: 286420

Lat: 55.306N

Lon: 61.503E

Elev: 234

StdP: 98.54

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.2	(c) -26.2	(d) -32.3	(e) 0.2	(f) -29.0	(g) -29.3	(h) 0.3	(i) -25.8	(j) 10.3	(k) -8.2	(l) 9.3	(m) -7.7	(n) 2.5	(o) 330	(p) 0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	30.8	19.2	28.9	18.7	27.0	18.0	21.0	27.4	20.0	26.2	19.1	24.9	4.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	14.1	23.1	17.9	13.3	22.6	17.0	12.5	21.6	61.5	27.5	58.3	26.2	55.2	24.8	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.2	9.0	7.9	-32.3	33.3	3.2	2.0	-34.6	34.7	-36.5	35.9	-38.3	37.0	-40.7	38.4
			-32.4	22.6	3.2	1.4	-34.7	23.6	-36.5	24.4	-38.3	25.2	-40.6	26.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-13.9	-12.0	-4.5	5.3	12.6	17.4	19.0	17.1	11.3	4.3	-5.0	-11.0		
	DBStd	12.95	6.74	6.72	4.91	5.46	5.17	4.35	3.73	4.22	4.53	5.02	6.93	6.80		
	HDD10.0	3326	742	615	449	161	33	3	0	1	36	187	449	650		
	HDD18.3	5576	1001	848	707	393	188	68	38	75	214	436	699	908		
	CDD10.0	943	0	0	0	18	114	224	279	222	76	9	0	0		
	CDD18.3	151	0	0	0	1	11	39	58	38	4	0	0	0		
	CDH23.3	1544	0	0	0	9	147	402	561	371	55	0	0	0		
	CDH26.7	455	0	0	0	1	29	122	169	121	13	0	0	0		
(14) Wind	WSAvg	4.1	3.6	3.7	4.5	4.7	4.5	4.2	3.7	3.7	4.0	4.3	4.1	3.8		
(15) Precipitation	PrecAvg	441	19	15	14	24	44	59	89	57	35	33	25	23		
	PrecMax	647	72	52	53	64	91	114	184	151	143	76	52	46		
	PrecMin	239	1	1	0	0	0	6	25	7	0	4	2	1		
	PrecStd	83	14	11	10	18	24	28	40	33	27	21	12	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.0	3.9	10.0	24.8	29.8	32.4	33.1	32.8	28.8	19.9	9.2	3.9		
		MCWB	0.4	1.1	5.7	13.6	17.4	18.7	20.3	20.4	16.5	11.5	6.2	1.6		
	2%	DB	-0.4	1.8	6.0	20.2	27.1	30.0	30.2	29.9	24.8	16.8	6.8	1.3		
		MCWB	-1.7	-0.2	2.7	11.1	15.5	18.7	19.5	19.4	15.0	10.3	4.6	-0.3		
	5%	DB	-2.2	-0.2	4.0	17.2	24.9	27.9	28.2	27.2	21.3	14.0	4.9	0.1		
		MCWB	-3.3	-1.8	1.4	9.7	14.7	18.1	19.2	18.8	14.1	8.9	3.1	-1.2		
	10%	DB	-4.1	-2.2	2.2	14.1	22.1	25.7	26.7	25.0	18.9	11.6	3.1	-1.8		
		MCWB	-5.0	-3.6	0.1	8.1	13.5	17.2	18.6	17.6	12.9	7.9	1.5	-2.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	1.6	5.7	14.0	18.6	21.4	22.2	22.1	18.0	12.7	6.5	1.9		
		MCDB	1.7	3.2	9.5	23.3	26.4	27.8	28.0	29.6	25.2	17.5	8.3	3.1		
	2%	WB	-1.7	-0.3	3.1	11.7	16.7	20.2	21.1	20.6	16.0	10.9	4.7	0.1		
		MCDB	-0.7	1.4	5.5	18.9	24.3	27.0	27.3	27.2	22.1	15.1	6.3	1.0		
	5%	WB	-3.3	-1.8	1.6	10.2	15.4	19.2	20.2	19.4	14.8	9.5	3.1	-1.1		
		MCDB	-2.4	-0.3	3.2	16.3	22.6	25.5	26.0	25.5	20.4	13.3	4.5	0.1		
	10%	WB	-5.1	-3.6	0.4	8.6	14.1	18.0	19.4	18.3	13.5	8.1	1.6	-2.8		
		MCDB	-4.2	-2.3	2.1	13.5	21.2	23.8	24.8	23.6	18.1	11.0	2.8	-1.8		
(35) Mean Daily Temperature Range	MDBR		7.7	8.8	8.6	10.2	11.8	10.9	10.6	10.2	9.8	7.4	6.5	7.1		
	5% DB	MCDBR	7.7	8.5	9.2	14.8	15.1	14.0	13.6	13.8	13.8	11.8	7.8	7.8		
		MCWBR	6.9	7.2	6.8	7.9	6.9	5.8	5.5	5.6	6.6	6.6	6.1	6.8		
	5% WB	MCDBR	7.4	8.0	8.4	13.7	13.5	12.3	11.8	12.5	12.7	10.6	7.0	7.8		
		MCWBR	6.8	7.0	6.6	7.8	7.0	5.9	5.6	5.8	6.8	6.5	5.9	6.8		
(40) Clear-Sky Solar Irradiance	taub		0.282	0.311	0.370	0.408	0.407	0.404	0.431	0.444	0.384	0.354	0.309	0.280		
	taud		2.069	2.047	1.987	2.125	2.233	2.312	2.227	2.163	2.296	2.255	2.161	2.044		
	Ebn at Noon		610	735	772	807	835	842	806	764	766	688	575	508		
	Edh at Noon		66	98	133	135	128	120	128	129	99	80	60	53		
(44) All-Sky Solar Radiation	RadAvg		0.85	1.92	3.42	4.38	5.20	5.65	5.48	4.28	2.89	1.45	0.73	0.52		
	RadStd		0.08	0.11	0.24	0.45	0.50	0.51	0.39	0.42	0.23	0.25	0.08	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page